Education

2024 – Present

PhD, Computing

University of Michigan–Flint · Audio Deepfake Detection, Signal Processing

2022 – 2024

PhD, Computer Science

Oakland University, Rochester, MI · Audio Deepfake Detection

2017 – 2020

MS, Computer Science

University of Engineering & Technology, Taxila, Pakistan · Thesis: Blur Image Region Detection and Segmentation

2013 – 2017

BS, Computer Science

University of Wah, Pakistan · Thesis: Performance Analysis of 802.11 Protocol

Experience

2026 – 2027

AI Developer Intern

ProbeTruth, Inc. · Flint, Michigan

- Designed, developed, and tested AI/ML models for real-world applications using Python and modern ML frameworks
- Built and optimized data pipelines for preprocessing, feature extraction, and model training
- Worked with multimodal data (text, audio, image, video) for AI-driven detection solutions

2024 – 2026

Research Assistant

University of Michigan–Flint · SMILES Lab

- Interpretable deepfake detection using symbolic and feature-based learning
- ML pipelines for large-scale audio datasets
- NSF-funded projects on multimodal deepfake analysis

2017 – 2020

Research Assistant (fully funded)

University of Engineering & Technology, Taxila, Pakistan

- Machine learning and computer vision research: feature design and statistical modeling
- Developed the Local Directional Mean Pattern (LDMP) feature extraction method

Publications

19. **Dual-Branch Architecture for Audio Deepfake Detection.**
    A. Khan, K. M. Malik. *INTERSPEECH (Main Track), 2026.* first author accepted

18. **TRACE: Training-Free Partial Audio Deepfake Detection via Embedding Trajectory Analysis of Speech Foundation Models.**
    A. Khan, K. M. Malik. *CVPR (MMFM), 2026.* first author accepted

17. **PSA-Net: Parallel Stacked Aggregated Network for Voice Authentication in IoT-Enabled Smart Devices.**
    A. Khan, I. U. Haq, K. M. Malik. *IEEE Trans. Consumer Electronics, 2026.* first author accepted

16. **Interpretable and Generalizable Audio Deepfake Detection using Joint Spectro-Temporal and Neuro-symbolic Learning.**
    A. Khan, K. M. Malik. *IEEE Trans. Information Forensics & Security, 2026.* first author under review

15. **Adversarial Attacks on Audio Deepfake Detection: A Benchmark and Comparative Study.**
    K. Uddin, M. U. Farooq, **A. Khan**, K. M. Malik. *arXiv:2509.07132, 2025.*

14. **SHIELD: A Secure and Highly Enhanced Integrated Learning for Robust Deepfake Detection Against Adversarial Attacks.**
    K. Uddin, **A. Khan**, M. U. Farooq, K. M. Malik. *arXiv:2507.13170, 2025.*

13. **Generalized Deepfake Detection Using Identity, Behavioral, and Geometric Signatures.**
    M. U. Farooq, **A. Khan**, I. U. Haq, K. M. Malik. *IEEE Trans. Computational Social Systems, 2025.*

12. **Transferable Adversarial Attacks on Audio Deepfake Detection.**
    M. U. Farooq, **A. Khan**, K. Uddin, K. M. Malik. *arXiv:2501.11902, 2025.*

11. **Securing Social Media Against Deepfakes Using Identity, Behavioral, and Geometric Signatures.**
    M. U. Farooq, **A. Khan**, I. U. Haq, K. M. Malik. *arXiv:2412.05487, 2024.*

10. **Bridging the Spoof Gap: A Unified Parallel Aggregation Network for Voice Presentation Attacks.**
    A. Khan, K. M. Malik. *Engineering Applications of AI · IF 8.5, 2023.* first author in process

9. **Securing Voice Biometrics: One-Shot Learning Approach for Audio Deepfake Detection.**
    A. Khan, K. M. Malik. *IEEE WIFS, 2023.* first author

8. **SpoTNet: A Spoofing-Aware Transformer Network for Effective Synthetic Speech Detection.**

A. Khan, K. M. Malik. *ACM-MAD, 2023.* first author

7. **Frame-to-Utterance Convergence: A Spectra-Temporal Approach for Unified Spoofing Detection.**
   A. Khan, K. M. Malik, S. Nawaz. *IEEE ICASSP, 2023.* first author

6. **Battling Voice Spoofing: A Review, Comparative Analysis, and Generalizability Evaluation of State-of-the-Art Voice Spoofing Countermeasures.**
   A. Khan, K. M. Malik, J. Ryan, M. Saravanan. *Artificial Intelligence Review, 56(Suppl. 1) · IF 12.0, 2023.* first author

5. **Toward Realigning Automatic Speaker Verification in the Era of COVID-19.**
   A. Khan, A. Javed, K. M. Malik, M. A. Raza, J. Ryan, A. K. J. Saudagar, H. Malik. *Sensors, 22(7), 2638 · IF 3.85, 2022.* first author

4. **A Robust Approach for Blur and Sharp Regions' Detection Using Multisequential Deviated Patterns.**
   A. Khan, A. Javed, A. Irtaza, M. T. Mahmood. *International Journal of Optics · Q2, 2021.* first author

3. **Defocus Blur Detection Using Novel Local Directional Mean Patterns (LDMP) and Segmentation via KNN Matting.**
   A. Khan, A. Irtaza, A. Javed, T. Nazir, H. Malik, K. M. Malik, M. A. Khan. *Frontiers of Computer Science, 16(2) · Q1, 2020.* first author

2. **Segmentation of Defocus Blur Using Local Triplicate Co-Occurrence Patterns (LTCoP).**
   A. Khan, S. A. Irtaza, A. Javed, M. A. Khan. *MACS-13, 2019.* first author

1. **Detection of Blur and Non-Blur Regions Using Frequency-Based Multi-Level Fusion Transformation and Classification via KNN Matting.**
   M. A. Khan, S. A. Irtaza, **A. Khan**. *MACS-13, 2019.*

Full list with links on the Publications page.

Skills

PROGRAMMING

Python, MATLAB, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

AUDIO & SIGNAL

Librosa, Praat, spectral analysis, handcrafted feature extraction

ML / DL

CNN, RNN, and transformer architectures for model design, training, and evaluation

TOOLS

Git, LaTeX, Jupyter, Google Colab, virtual machines

Awards & Scholarships

- Fully Funded Research Scholar · University of Michigan–Flint
- HEC Pakistan PhD Scholarship · Higher Education Commission, Pakistan
- Fully Funded Master's Scholarship · UET Taxila
- Appreciation Award · MACS-13 Conference
- 1st Position · PBTE Lahore board examinations

Referees

Dr. Khalid Mahmood Malik

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